

String Transform

Time limit: 1.00 s

Memory limit: 512 MB

Consider the following string transformation:

1. append the character # to the string (we assume that # is lexicographically smaller than all other characters of the string)
2. generate all rotations of the string
3. sort the rotations in increasing order
4. based on this order, construct a new string that contains the last character of each rotation

For example, the string `bab` becomes `bab#`. Then, the sorted list of rotations is `#bab`, `abc#b`, `bab`, `bc#ba`, and `c#bab`. This yields a string `cb#ab`.

Input

The only input line contains the transformed string of length $n + 1$. Each character of the original string is one of a–z.

Output

Print the original string of length n .

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
cb#ab	bab